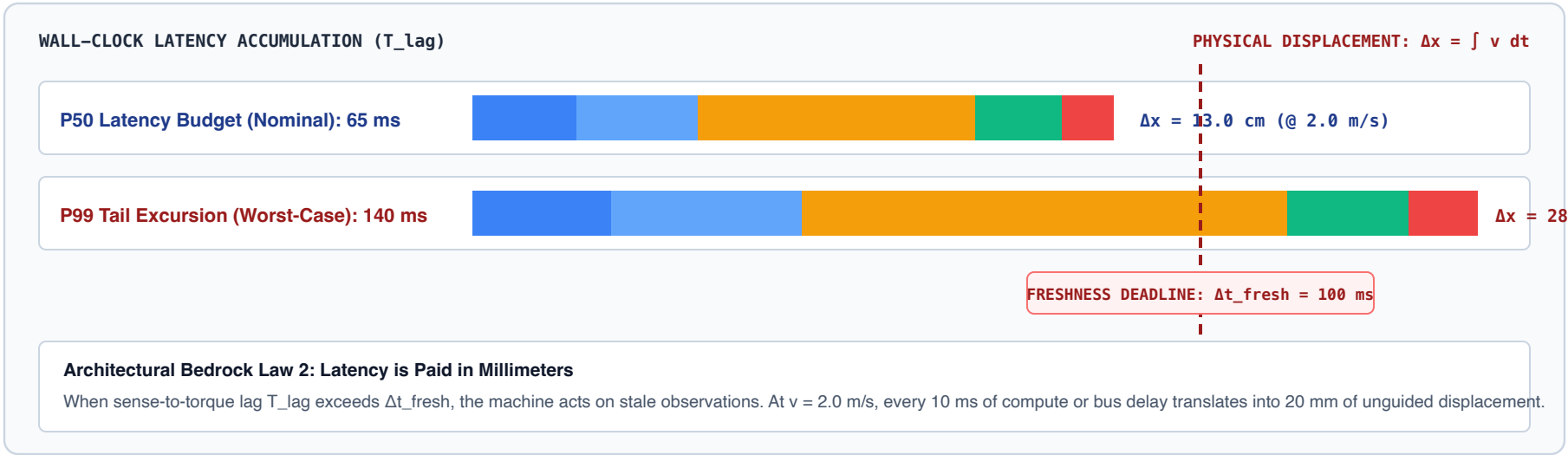
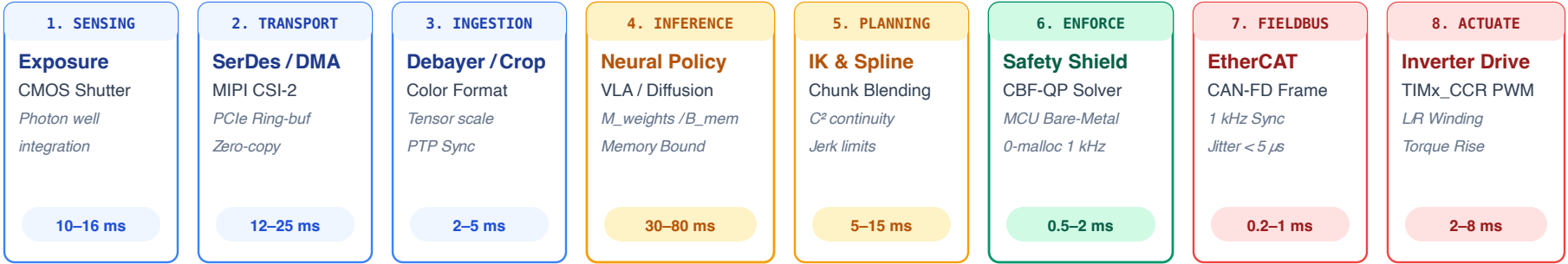


END-TO-END SENSE-TO-ACTUATION LATENCY WATERFALL & FRESHNESS BUDGET

Cumulative Wall-Clock Decomposition: From Photon Transduction to Torque Generation vs. the Freshness Deadline



End-to-end sense-to-actuation latency combines sensor integration, bus transport, policy inference, safety validation, and actuator electromagnetic rise.